15
Radiative collisions

A radiative collision is a resonant energy transfer collision in which two atoms absorb or emit photons during the collision.[1] Alternatively, a radiative collision is the emission or absorption of a photon from a transient molecule, and, as shown by Gallagher and Holstein,[2] radiative collisions can also be described in terms of line broadening. In a line broadening experiment there are typically many atoms and weak radiation fields, and in a radiative collision experiment there are few atoms and intense radiation fields. The only real difference is whether there are many atoms or many photons. Due to the short collision times, $\sim 10^{-12}$ s, simply observing radiative collisions between low lying states requires high optical powers, and entering the regime where the optical field is no longer a minor perturbation seems unlikely. Due to their long collision times and large dipole moments, Rydberg atoms provide the ideal system in which to study radiative collisions in a quantitative fashion. As we shall see, it is straightforward to enter the strong field regime in which the radiation field, a microwave or rf field to be precise, is no longer a minor perturbation. Ironically, while the experiments are radiative collision experiments, with few atoms and many photons, the description of the strong field regime is given in terms of dressed molecular states, which is more similar to a line broadening description.[3]

Using a simple, three level model we can develop a feeling for the microwave powers required to observe radiatively assisted collisional energy transfer between Rydberg atoms.[3] Consider the dipole–dipole atomic system shown in Fig. 15.1(a). In the Na ns + ns→np + $(n-1)$p resonant collisions described in the previous chapter the ns state corresponds to both s and s′ of Fig. 15.1(a) and the $n-1$ and np states correspond to p and p' of Fig. 15.1(a), respectively. The collisions occurs via the interaction

$$V = \frac{\mu_1\mu_2}{r^3}. \tag{15.1}$$

In Fig. 15.1(b) we show the level system of a collision radiatively assisted by the absorption of one photon of frequency ω. In this case the interaction leading to the production of the p and d states is given by

$$V = \frac{\mu_1\mu_2\mu_3\,E}{r^3\Delta}, \tag{15.2}$$

where Δ is the detuning between the real and virtual intermediate p' states and E is

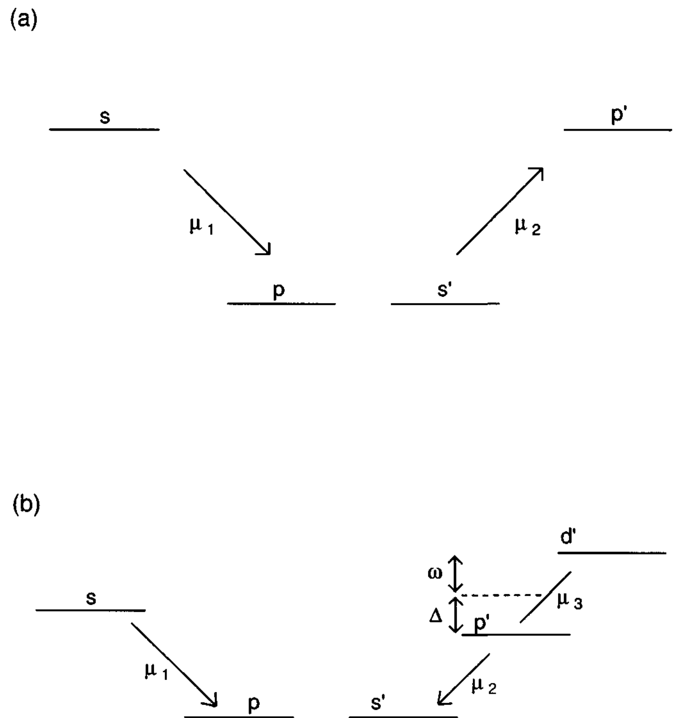

Fig. 15.1 (a) Energy levels and dipole matrix elements for the resonant collision of two atoms in the s and s′ states resulting in the production of two atoms in the p and p′ states. (b) Energy levels and matrix elements for the radiative collision in which an s and an s′ atom collide to produce atoms in the p and d′ states. The production of the d′ state is via the virtual p′ state which is detuned from the real p′ state by an energy Δ.

the amplitude of the microwave field in which the collision occurs. The expressions of Eqs. (15.1) and (15.2) differ by the factor $\mu_3 E/\Delta$. For the radiatively assisted collision shown in Fig. 15.1(b) to be ~10% as probable as the resonant collision of Fig. 15.1a, we simply require that

$$\frac{\mu_3 E}{\Delta} = \frac{1}{3}. \tag{15.3}$$

If we assume typical values of $\mu_3 \simeq n^2$ and $\Delta \sim 0.1n^{-3}$, the required field is given by

$$E = \frac{1}{30n^5}. \tag{15.4}$$

Evaluating Eq. (15.4) at $n = 20$ leads to a field of 50 V/cm, or a power of ~6 W/cm^2, a power roughly six orders of magnitude lower than those used in optical radiative collision experiments with low lying atomic states.[4–7] These powers are not only readily obtained, but are easily exceeded by orders of magnitude, so that

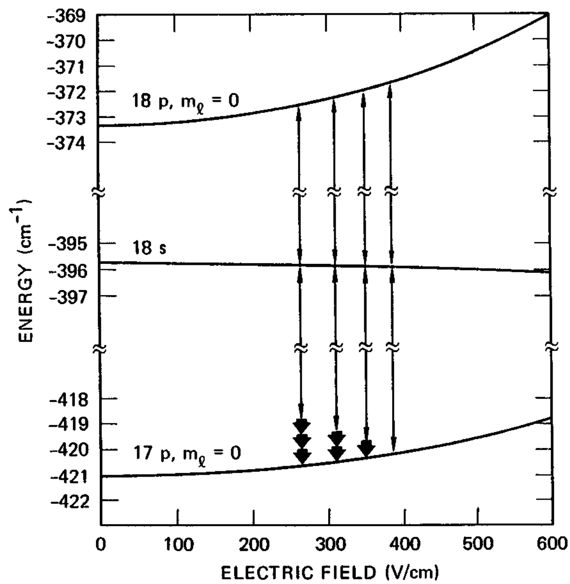

Fig. 15.2 Stark energy level diagram of the Na $|m_\ell| = 0$ states, relevant to the multiphoton-assisted collisions. The vertical lines indicate the collisional transfer and are drawn at the fields where they occur. The thick arrows correspond to the emitted photons (from ref. 8).

it is a straightforward matter to reach the strong field regime, in which the effect of the field is no longer a minor perturbation.[8]

The Na ns + Na $ns \rightarrow$ Na np + Na $(n-1)$p collisions in combined static and microwave fields were the first Rydberg atom radiative collisions studied. Specifically, the process

$$\text{Na 18s} + \text{Na 18s} \rightarrow \text{Na 18p} + \text{Na 17p} + m\omega \qquad (15.5)$$

has been examined in detail. Here m is the number of photons emitted in the collision. When $m > 0$ there is emission, and when $m < 0$ there is absorption. The energy levels of the Na 18s, 17p, and 18p states are shown as a function of static field in Fig. 15.2.[8] For simplicity we have only shown the $m_\ell = 0$ levels in Fig. 15.2. The (0,0) resonant collision, with no microwaves, occurs at a field of 390 V/cm, and the radiative collisions accompanied by the stimulated emission of one, two, and three photons occur at lower static fields, as shown. We follow the convention of labelling the collisional resonances by the m_ℓ values of the lower and upper final p states of the collision, i.e. by $(m_\mathrm{l},m_\mathrm{u})$, where m_l and m_u are the m_ℓ values of the lower and upper p states. The number of photons emitted is shown by the number of short bold arrows. In Fig. 15.2 the field displacement of a radiative collision resonance from the normal collisional resonance is proportional to the microwave frequency and inversely proportional to the sum of the differential Stark shifts of the p states. As shown in Fig. 15.2, the 18s state has a very small Stark shift, which we ignore. In principle, it is equally possible to observe collisions in which there is absorption, as opposed to emission, of several microwave photons. However, in the Na ns + Na $ns \rightarrow$ Na np + Na $(n-1)$p system these radiatively assisted collisions occur at higher static fields, where there are many collisional resonances due to the Stark states which are not shown in Fig. 15.2.[9] The n = 17 and n=16 Stark manifolds lie just below the 18p and 17p states, and at fields ~10% higher than the fields at which the resonant collisions of Fig. 15.2 occur, resonant collisions involving the Stark states appear. The presence of these collisional resonances precludes the unambiguous observation of $ns + ns \rightarrow n\mathrm{p} + (n-1)p$ collisional resonances in which microwave photons are absorbed.

Initial experimental study of radiative collisions

The initial experimental study of radiative collisions between Rydberg atoms was done by simply introducing a microwave field into the region between the two field plates of an apparatus such as the one shown in Fig. 14.7.[10] The microwave field was introduced using a horn outside the plates, and several microwave frequencies between 12 and 18 GHz which propagated well between the plates were used. The maximum intensities were estimated to be ~3 W/cm^2. The data are accumulated in the same fashion as in the resonant collision experiment, the only difference being the presence of a microwave field. The atoms are excited by two pulsed lasers, and collisions are allowed to occur for a 1 μs period, after which a high voltage pulse is applied to the lower field plate to field ionize atoms in the np state. The signal from the np state atoms is recorded as the static field is scanned. On each scan the microwave field amplitude is held constant. The dependence of the radiative collision cross section on the microwave power is determined by repeating the static field scan at different microwave powers. In these experiments single photon radiative collisions were observed, at static fields slightly below the fields at which the resonant collisions were observed, as expected from Fig. 15.2. A typical example is shown in Fig. 15.3. The field displacement of the radiative collision signals from the resonant collision signal is equal to the field required to produce a Stark shift equal to the microwave frequency. Two points about this experiment are worth noting. First, the radiatively assisted collision cross sections are within an order of magnitude of the resonant collision cross sections for incident microwave intensities of a few W/cm^2, confirming the estimate of Eq. (15.4). Second, it is interesting to observe that at low microwave power the radiatively assisted collision signal is linear in the microwave power, but at higher powers the radiatively assisted collision signal rises less rapidly, suggesting the entrance into a nonperturbative regime.

Replacing the field plates of Fig. 14.7 with the resonant microwave cavity shown in Fig. 15.4 allowed an increase in the circulating microwave power by Q, the quality factor, of the cavity.[8] The cavity is a piece of WR-90 (X band) waveguide 20 cm long which is closed at both ends. The inside dimensions of the cavity are

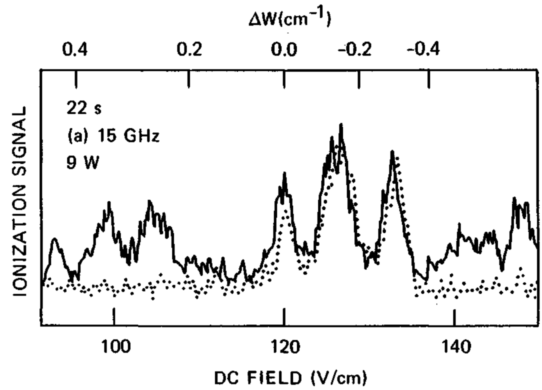

Fig. 15.3 The observed Na 22p ion signal after the population of 22s state vs the dc field in the presence of a 15 GHz microwave field (solid trace). The dotted trace was observed with no microwave field, and the sharp resonances in the center (for both solid and dotted traces) result from the resonant collisions, while the displaced resonances on the side (solid trace) are due to microwave assisted collisions. The scale on the top of the figure shows the detuning from the (0,0) resonance (from ref. 10).

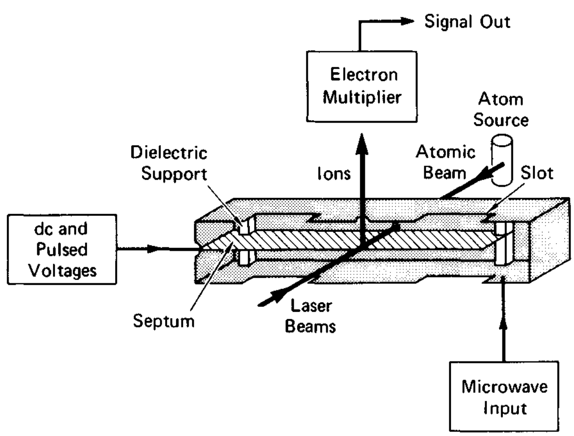

Fig. 15.4 Main features of the atomic beam apparatus, the atomic source, the microwave cavity, and the electron multiplier. The microwave cavity is shown sliced in half. The Cu septum bisects the height of the cavity. Two holes of diameter 1.3 mm are drilled in the sidewalls to admit the collinear laser and Na atomic beams, and a 1 mm diameter hole in the top of the cavity allows Na^+ resulting from field ionization of Na to be extracted. Note the slots for pumping (from ref. 8).

20.32 cm long, 1.02 cm high, and 2.28 cm wide. The cavity contains a septum, as shown in Fig. 15.4, which allows the application of a static tuning field and a pulsed field in the vertical direction. The presence of the septum does not affect the TE_{10n} cavity modes since they have only vertical electric fields.

The cavity is operated on modes of odd n, which have an electric field antinode at the center of the cavity. Specifically the $n = 15, 17$, and 19 modes, with resonant frequencies of 12.8, 14.4, and 15.5 GHz have been used. These modes have $Q \sim 1300$, and when excited with a power of 20 W, the circulating microwave intensity is ~10 kW/cm^2, and the microwave field is ~500 V/cm. As shown in Fig. 15.4, the atomic and laser beams enter the cavity through opposing holes in the sidewalls of the cavity ~1 mm above the septum. Thus a cylindrical volume of Rydberg atoms is produced. There is a small, ~1 mm diameter, hole in the center of the top of the cavity through which ions produced by field ionization of atoms directly beneath the hole escape from the cavity en route to the detector. Since the atoms only travel ~1 mm during the time between laser excitation and field ionization, the location of the hole in the top ensures that the signals observed are due to atoms at an antinode of the microwave electric field.

Data are accumulated in the same way as used to acquire those shown in Fig. 15.3. Not surprisingly, the radiative collisions which had previously been observed with 10 W of microwave power can be observed with milliwatts of power, due to the high Q of the cavity. The most interesting aspect of the data, though, is what happens in strong microwave fields. In Fig. 15.5 we show the development of the resonances observed for the radiative collision process of Eq. (15.5) as the 15.5 GHz microwave power incident on the cavity is increased.[8] With no microwave power only the four collisional resonances corresponding to the different $|m|$ values of the 17p and 18p states are observed. When the microwave power is raised this pattern of four resonances is repeated three times at lower static tuning fields, corresponding to the stimulated emission of one, two, and three microwave photons during the collision. A straightforward way of labeling the resonances is simply to extend our previous notation to account for the number of photons emitted. Explicitly, we label the collisional resonances as $(m_l,m_u)^m$ where m_l and m_u are the m_ℓ values of the upper and lower p states and m is the number of photons emitted. In Fig. 15.5 the $(0,0)^m$ resonances are indicated by m with an arrow.

It is hardly surprising that, as the microwave power is raised, higher order multiphoton processes are observed. On the other hand, it may be surprising that for each $m \neq 0$ the cross sections first increase then decrease with microwave power. For example, the $m = 1$ cross section is clearly zero in the lowest trace. Similarly, the $m = 0$ cross section vanishes in the trace one above the lowest but reappears in the lowest trace. Such behavior, typical of the strong field regime, is not predicted by perturbation theory. Close inspection of Fig. 15.5 reveals that the positions of the collisional resonances shift to lower static field as the microwave power is raised. Finally, in contrast to the usual observation of broadening with increased power, the $(0,0)^m$ resonances, which are well isolated from other resonances, develop from broad asymmetric resonances to narrow symmetric ones as the microwave power is raised.

That the observed resonances for $m \neq 0$ become narrower and more symmetric as the power is increased can be understood in the following way, using the one-photon process as an example. The coupling matrix element of Eq. (15.2) has two

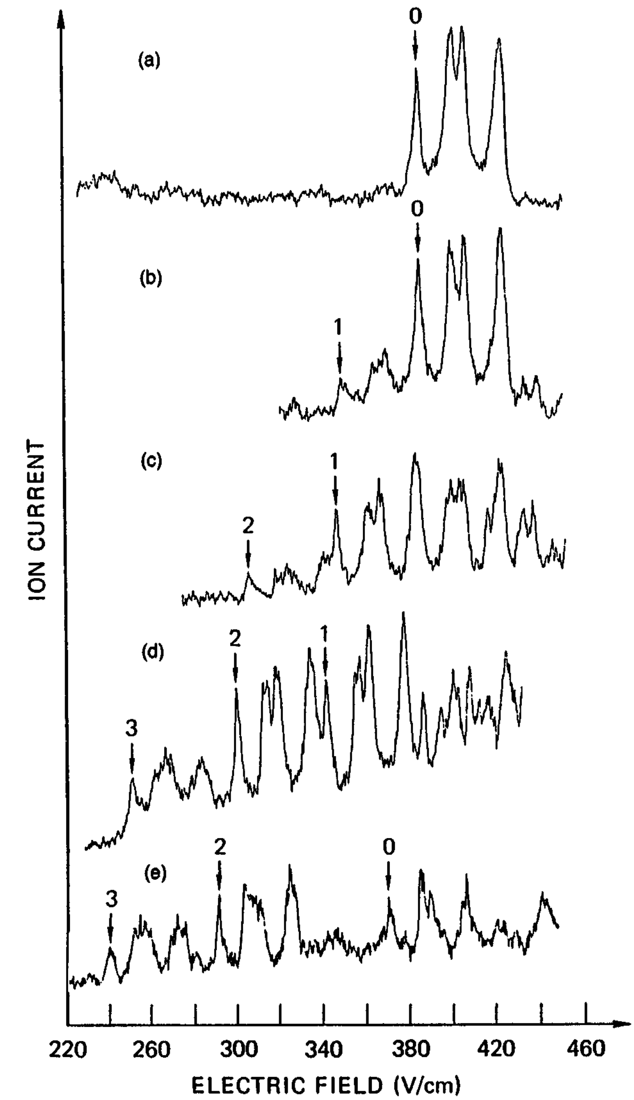

Fig. 15.5 Observed Na 18p ion signal after the population of the 18s level vs the static field with a 15.4 GHz microwave field. Trace (a) corresponds to no microwave power input to the cavity and shows the set of four zero-photon collisional resonances. Traces (b), (c), (d), and (e) correspond, respectively, to 13.5, 50, 105, and 165 V/cm microwave field amplitudes inside the cavity and show additional sets of four collisional resonances corresponding to one, two, and three-photon radiatively assisted collisions. The peaks labelled 0, 1, 2, and 3 correspond to the lowest field member of the set of four resonances corresponding to zero-, one-, two-, and three-photon assisted collisions, $(0,0)^0$, $(0,0)^1$, $(0,0)^2$, $(0,0)^3$ (from ref. 8).

factors, a dipole–dipole term, and a dipole–field term. For the collision to be observable, the time integral of the matrix element over the collision must exceed some minimum value ~1. Thus, if the microwave field is smaller, the radiative collision must occur with a smaller impact parameter to compensate. A small impact parameter has two effects. First, it implies a short collision time and a wide resonance. Second, as the atoms come closer together the molecular energy levels deviate from their $r = \infty$ values, producing the asymmetry. As the microwave field is raised, the impact parameter can become larger, and the resonances become narrower and more symmetric.

Theoretical description

The experimental data shown in Figs. 15.3 and 15.5 were obtained with a microwave frequency $\omega/2\pi > 1/\tau$ where τ is the time or duration of the collision and $1/\tau$ is the linewidth. In this case the resonances corresponding to the absorption or emission of different numbers of photons are resolved. Here we describe radiative collisions starting from the high frequency regime, $\omega/2\pi > 1/\tau$ and progressing to the low frequency regime, $\omega/2\pi < 1/\tau$.

The strong field, high frequency regime

To describe the shifts and intensities of the m-photon assisted collisional resonances with the microwave field Pillet *et al.* developed a picture based on dressed molecular states,[3] and we follow that development here. As in the previous chapter, we break the Hamiltonian into an unperturbed Hamiltonian H_0, and a perturbation V. The difference from our previous treatment of resonant collisions is that now H_0 describes the isolated, noninteracting, atoms in both static and microwave fields. Each of the two atoms is described by a dressed atomic state, and we construct the dressed molecular state as a direct product of the two atomic states. The dipole–dipole interaction V is still given by Eq. (14.12), and using it we can calculate the transition probabilities and cross sections for the radiatively assisted collisions.

We begin by developing the wavefunction for one atom in combined static and microwave fields, both of which are in the same direction. If the nominal $n\ell$ state has an energy $W_{n\ell}(E)$ in a static field E, we can describe the energy in the vicinity of the static field E_S by

$$W_{n\ell}\,(E_S+\Delta E) = W'_{n\ell} + k_{n\ell}\Delta E + \frac{1}{2}\alpha_{n\ell}(\Delta E)^2 \qquad (15.6)$$

where

$$W'_{n\ell} = W_{n\ell}(E_S), \quad \mathrm{k}_{n\ell} = \mathrm{d}\frac{W_{n\ell}}{\mathrm{d}E}\bigg|_{E_S}, \quad \text{and} \quad \alpha_{n\ell} = \frac{\mathrm{d}^2 W_{n\ell}}{\mathrm{d}E^2}\bigg|_{E_S}$$

If we now imagine that ΔE varies in time, but slowly, its only effect is to cause a time variation of the energy of the $n\ell$ state. We assume that the spatial wavefunction is unaffected by ΔE and that no transitions occur. This approximation is the adiabatic approximation of Autler and Townes.[11] Now let us consider the time variation of ΔE of particular interest to us, $E_{\mathrm{mw}} \cos \omega t$. If the assumptions stated above are valid, we can use the energy of Eq. (15.6) as the unperturbed Hamiltonian H_0 in the Schroedinger equation. Explicitly,

$$W(t)\,\psi_{n\ell}(t) = (W'_{n\ell} + k_{n\ell}\,E_{\mathrm{mw}} \cos \omega t + \frac{\alpha_{n\ell}}{2}(E_{\mathrm{mw}} \cos \omega t)^2)\psi_{n\ell}(t) = \frac{\mathrm{i}\partial \psi_{n\ell}(t)}{\partial t}, \tag{15.7}$$

which has the solution

$$\psi_{n\ell}(t) = \psi_{n\ell}\,\mathrm{e}^{-\mathrm{i}\int_{t_0}^{t} W(t')\,\mathrm{d}t'}. \tag{15.8}$$

Carrying out the integrations and discarding the phase factors from t_0 yields

$$\psi_{n\ell}(t) = \psi_{n\ell}\,\mathrm{e}^{-\mathrm{i}\left(W' + \frac{\alpha_{n\ell}}{4}\right)t}\,\mathrm{e}^{-\mathrm{i}\frac{k_{n\ell}E_{\mathrm{mw}}}{\omega}\sin \omega t}\,\mathrm{e}^{-\mathrm{i}\frac{\alpha_{n\ell}E_{\mathrm{mw}}^2}{8\omega}\sin 2\omega t}. \tag{15.9}$$

The exponentials with sinusoidal arguments may be expanded as Bessel functions using[12]

$$\mathrm{e}^{\mathrm{i}x \sin \omega t} = \sum_{-\infty}^{\infty} J_k(x)\,\mathrm{e}^{\mathrm{i}k\omega t}. \tag{15.10}$$

With the Bessel function expansion of Eq. (15.10), Eq. (15.9) becomes

$$\psi_{n\ell}(t) = \psi_{n\ell}\,\mathrm{e}^{-\mathrm{i}\left(W'_{n\ell} + \frac{\alpha_{n\ell}E_{\mathrm{mw}}^2}{4}\right)t}\left(\sum_{-\infty}^{\infty} \mathrm{J}_{\mathrm{k}}\left(\frac{\mathrm{k}_{n\ell}{}'E_{\mathrm{mw}}}{\omega}\right)\mathrm{e}^{-\mathrm{i}k\omega t}\right)\left(\sum_{\infty}^{\infty} J_{k'}\left(\frac{\alpha_{n\ell}E_{\mathrm{mw}}^2}{8\omega}\right)\mathrm{e}^{-\mathrm{i}2k'\omega t}\right). \tag{15.11}$$

If $\alpha_{n\ell}E_{\mathrm{mw}}{}^2 \ll \omega$, as is usually the case for the experiments described here, in the last expansion of Eq. (15.11) only the $k' = 0$ term contributes and the summation is equal to one. In this case the wavefunction is written as

$$\psi_{n\ell}(t) = \psi_{n\ell}\,\mathrm{e}^{-\mathrm{i}\left(W'_{n\ell} + \frac{\alpha_{n\ell}E_{\mathrm{mw}}^2}{4}\right)t}\sum_{-\infty}^{\infty} J_k\left(\frac{k_{n\ell}E_{\mathrm{mw}}}{\omega}\right)\mathrm{e}^{-\mathrm{i}k\omega t}. \tag{15.12}$$

We note that for collisions involving states of good parity, $k_{n\ell} = 0$, and the first summation of Eq. (15.11) collapses to 1. In this case we may not disregard the second summation of Eq. (15.11).

As discussed in Chapter 10, we can think of the effect of the microwave field on the $n\ell$ state as modulating its energy. Just as modulating the frequency of a radio wave produces sidebands, the microwave field can be thought of as modulating the energy of the $n\ell$ state and breaking it into carrier and sideband states.

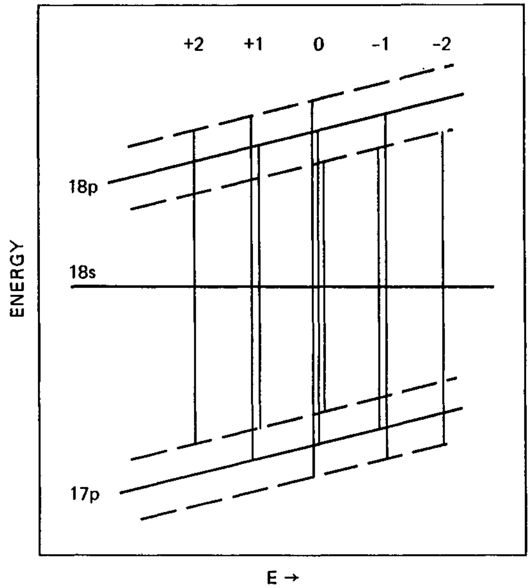

Fig. 15.6 Energy levels of the Na 17p, 18s, and 18p states showing the first upper and lower sideband states of the p states. The numbers +2,. . ., −2 refer to the net number of photons emitted in a radiative collision at these tunings of the field. Note that there are several processes which lead to the net emission of, for example, zero photons (from ref. 3).

In Eq. (15.12) $\psi_{n\ell}$ is the spatial wavefunction of the $n\ell$ state at the static field E_S, $W'_{n\ell}$ is the energy in the static field E_S, $\alpha_{n\ell}E_{mw}{}^2/4$ is the ac Stark shift produced by the microwave field, and $W'_{n\ell} + \alpha_{n\ell}E_{mw}{}^2/4$ is the carrier energy. The Bessel function $J_k(kE_{mw}/\omega)$ gives the amplitude in the kth sideband state, which is displaced in energy from the carrier state by $k\omega$, k microwave photons. For the np states $k_{n\mathrm{p}}$ is large, but for the ns states $k_{n\mathrm{s}}$ is very small, and the amplitudes of the s sidebands are negligible.

A graphic presentation of the effect of the microwave fields is shown in Fig. 15.6, a plot of the energies of the 18s, 17p and 18p states showing the carrier energies of these states as well as the energies of the ±1 sideband states of the *p* states.[3] As mentioned above, and shown explicitly in Eq. (15.12), the amplitude of each sideband depends on the strength of the microwave field, and while this point is not made explicitly in Fig. 15.6, it is implicit in that no sidebands of the *s* state are shown and second and higher sidebands of the *p* states are not shown. Thus Fig. 15.6 corresponds to a microwave field too weak to have observable amplitudes for the second sidebands of the *p* states. With no microwaves all sidebands would have zero amplitude and the carriers amplitude one. A further interesting point to observe in Fig. 15.6 is that there are two processes corresponding to the net emission of ±1 photon, and three corresponding to the net emission of zero photons. These processes correspond to the transitions between the energetically allowed combinations of sideband states, raising the question of how to account for these different processes, all of which occur at the same static field and are therefore experimentally indistinguishable.

Using the dressed atomic states we construct the dressed molecular states as direct products. Explicitly, we construct the two states $\psi_A(t)$ and $\psi_B(t)$ given by[3]

$$\psi_A(t) = \psi_{1ns}(t) \otimes \psi_{2ns}(t) \tag{15.13a}$$

and

$$\psi_B(t) = \psi_{1np}(t) \otimes \psi_{2(n\text{-}1)p}(t) \tag{15.13b}$$

in which each of the individual wavefunctions is given by Eq. (15.12). If we write out the expression for ψ_B explicitly we find

$$\psi_B(t) = \psi_{1np}\psi_{2(n-1)p}\, e^{-iW_B t} \sum_{kk'} J_k\left(\frac{k_{np}E_{mw}}{\omega}\right) J_{k'}\left(\frac{k_{(n-1)p}E_{mw}}{\omega}\right) e^{-i(k+k')\omega t}, \tag{15.14}$$

where $W_B = W'_{np} + W'_{(n-1)p} + (\alpha_{np} + \alpha_{(n-1)p})E^2_{mw}/4$, the sum of the carrier energies of the np and $(n-1)$p states. With the Bessel function identity,[13]

$$J_N(x \pm y) = \sum_{-\infty}^{\infty} J_{N\mp k}(x) J_k(y), \tag{15.15}$$

the double summation of Eq. (15.14) may be condensed to a single sum, yielding

$$\psi_B(t) = \psi_{1np}\psi_{2(n-1)p}\, e^{-iW_B t} \sum J_k\left(\frac{k_B E_{mw}}{\omega}\right) e^{-ik\omega t}, \tag{15.16}$$

where $k_B = k_{np} + k_{(n-1)p}$. Similarly we can write the wavefunction for $\psi_A(t)$ as

$$\psi_A(t) = \psi_{1ns}\psi_{2ns}\, e^{-iW_A t} \sum J_{k'}\left(\frac{k_A E_{mw}}{\omega}\right) e^{-ik'\omega t}, \tag{15.17}$$

where $W_A = 2W'_{ns}$ and $k_A = 2k_{ns}$

Both $\psi_A(t)$ and $\psi_B(t)$ are solutions to H_0, and the total wavefunction is, in general, a linear combination of the two. Explicitly, it is given by

$$\psi(t) = C_A(t)\psi_A(t) + C_B(t)\psi_B(t), \tag{15.18}$$

where $C_A(t)$ and $C_B(t)$ are time dependent coefficients analogous to those in Eq. (14.15). If we insert the wavefunction of Eq. (15.18) into the time dependent Schroedinger equation,

$$H\Psi(t) = i\partial\Psi/\partial t, \tag{15.19}$$

and use the fact that $\psi_A(t)$ and $\psi_B(t)$ are solutions of the unperturbed Schroedinger equation, we find the equation

$$VC_A(t)\psi_A(t) + VC_B(t)\psi_B(t) = i\,\dot{C}_A(t)\psi_A(t) + i\dot{C}_B(t)\psi_B(t). \tag{15.20}$$

Assuming that the diagonal matrix elements of V vanish, multiplying Eq. (15.20) in turn by $\psi^*_A(t)$ and $\psi^*_B(t)$ and carrying out the spatial integrations yields

$$\langle\psi_B(t)|V|\psi_A(t)\rangle C_A(t) = i\dot{C}_B(t) \tag{15.21a}$$

and

$$\langle\psi_A(t)|V|\psi_B(t)\rangle\, C_B(t) = \mathrm{i}\dot{C}_A(t). \tag{15.21b}$$

We note that Eqs. (15.21) are similar to Eqs. (14.16) but for the time dependence of $\psi_A(t)$ and $\psi_B(t)$. We recall that V is always time dependent since it depends on the internuclear separation of the atoms. However, this time dependence is slow compared to those of $\psi_A(t)$ and $\psi_B(t)$

Our interest is in computing the transition probability at resonance, a condition which is met when $W_A = W_B + m\omega$ where m is an integer. First, we explicitly write the matrix element $\langle\psi_B(t)|V|\psi_A(t)\rangle$ using the wavefunctions of Eqs. (15.16) and (15.17) and the Bessel function relation of Eq. (15.15). It is given by

$$\langle\psi_B(t)|V|\psi_A(t)\rangle = \langle\psi_B|V|\psi_A\rangle \sum_k \mathrm{J}_k\left(\frac{(k_A - k_B)\,E_{mw}}{\omega}\right) \mathrm{e}^{-\mathrm{i}[W_A - W_B + k\omega]t}. \tag{15.22}$$

In this form it is apparent that the matrix element $\langle\psi_B(t)|V|\psi_A(t)\rangle$ is a product of the spatial matrix element, which varies slowly with time as the atoms move past each other, and the Bessel function sum, which contains an enormous range of frequencies. At resonance one term of the sum is constant in time and the others oscillate at multiples of ω. The oscillating terms are ineffective in causing transitions in collisions of duration much longer than the microwave period, since they average to zero. Thus at the resonance corresponding to the radiative collision in which m photons are emitted, i.e. when $W_A = W_B + m\omega$, the constant and significant part of the matrix element of Eq. (15.22) is given by

$$\langle\psi_B(t)|V|\psi_A(t)\rangle = \langle\psi_B|V|\psi_A\rangle\, J_m\left(\frac{K\,E_{mw}}{\omega}\right) \tag{15.23}$$

where $K = k_A - k_B$. The matrix element of Eq. (15.23) is identical to the matrix element for a normal resonant collision multiplied by a constant factor, the Bessel function, and as a result the transition probability can be computed using the same approaches as in Chapter 14.

All radiative collision experiments with Rydberg atoms have been done with the collision velocity perpendicular to the static and microwave fields. Thus the results obtained are an average over spatial orientations, and it makes little sense to use a detailed model of the interaction. Accordingly, we assume that

$$|\langle\psi_B|V|\psi_A\rangle| = \begin{cases} \dfrac{\chi}{b^3} & r < \dfrac{\sqrt{5}b}{2} \\[2ex] 0 & r > \dfrac{\sqrt{5}b}{2} \end{cases} \tag{15.24}$$

where b is the impact parameter and $\chi = \mu_1\mu_2$. With this assumption, the atoms interact for a time b/v. Using matrix elements of the form of Eq. (15.24) leads to sinusoidally oscillating solutions of $C_A(t)$ and $C_B(t)$ during the collision . If we choose as initial conditions $C_A(-\infty)=1$ and $C_B(-\infty)=0$, then after the collision,

$$C_A^2(t)=\cos^2\left[\frac{\chi}{b^2v}J_m\left(\frac{KE_{mw}}{\omega}\right)\right], \tag{15.25a}$$

and the transition probability at impact parameter b, $P_m(b)$, is given by $C_B^2(t)$. Explicitly,

$$P_m(b)=C_B^2(t)=\sin^2\left[\frac{\chi}{b^2v}J_m\left(\frac{KE_{mw}}{\omega}\right)\right]. \tag{15.25b}$$

The transition probability of Eq. (15.25b) is of the same form as the transition probability of Eq. (20.25), with $D=\chi|J_m(KE_{mw}/\omega)|$. Accordingly, if we define an impact parameter $b_m(E_{mw})$ such that

$$\frac{\chi}{b_m^2(E_{mw})v}\left|J_m\left(\frac{KE_{mw}}{\omega}\right)\right|=\frac{\pi}{2}, \tag{15.26}$$

$P_m(b)$ oscillates between 0 and 1 as b increases from zero to $b_m(E_{mw})$, at which point $P_m(b)=1$ for the last time. As b increases from $b_m(E_{mw})$ to infinity $P_m(b)$ decreases monotonically from 1 to 0.

To obtain the cross section for the m-photon assisted collision in the presence of a microwave field E_{mw} we integrate the transition probability over impact parameter. Explicitly

$$\sigma_m(E_{mw})=\int_0^\infty 2\pi b\sin^2\frac{\chi}{b^2v}\left|J_m\left(\frac{KE_{mw}}{\omega}\right)\right|\mathrm{d}b. \tag{15.27}$$

Carrying out the integral explicitly yields

$$\sigma_m(E_{mw})=\frac{\pi^3}{4}b_m^2(E_{mw})=\frac{\pi^2}{2}\frac{\chi}{v}\left|J_m\left(\frac{KE_{mw}}{\omega}\right)\right|. \tag{15.28}$$

A radiative collision with the emission of no photons in the absence of a microwave field is simply a resonant collision. Since $J_0(0)=1$, we can express the radiative collision cross section of Eq. (15.28) as

$$\sigma_m(E_{mw})=\sigma_R\left|J_m\left(\frac{KE_{mw}}{\omega}\right)\right|, \tag{15.29}$$

i.e. as the product of the resonant collision cross section σ_R and the magnitude of the $J_m(KE_{mw}/\omega)$ Bessel function.

Eq. (15.29) is only valid in the strong field regime. It is clear that if $J_m(KE_{mw}/\omega)$ is small, using Eq. (15.26) we compute a small value of $b_m(E_{mw})$. When this value approaches the size of the atom, Eq. (15.29) is no longer valid, for the integral of Eq. (15.27) is dominated by contributions for $b\sim b_m(E_{mw})$, where neither the assumption of the energies being independent of r nor the dipole approximation are valid. Under such circumstances a collision with a small impact parameter does not lead to as large a transition probability as indicated by Eq. (15.25b).

Connection to the weak field regime

In the experiments with Rydberg atoms it is very difficult to observe radiatively assisted collisions with cross sections more than a factor of 10 smaller than the resonant collision cross sections, so the deviations from Eq. (15.29) are not apparent. However, in other contexts, such as laser assisted collisions, this limitation does not apply, and it is interesting to consider how the above description passes over into the weak field regime, in which $J_m(KE_{\mathrm{mw}}/\omega)$ is small. If we restrict the integration in Eq. (15.27) to the large r region of space, in which the approximations we have used are valid, we can rewrite Eq. (15.27) as

$$\sigma_m = \int_B^\infty \frac{\chi^2}{b^4 v^2} J_m^2\left(\frac{KE_{\mathrm{mw}}}{\omega}\right) 2\pi b\, \mathrm{d}b$$
$$= \frac{\pi\chi^2}{B^2 v^2} J_m^2\left(\frac{KE_{\mathrm{mw}}}{\omega}\right), \tag{15.30}$$

in which an obvious lower limit for B is the geometrical radius of the atom. Acccordingly, $B = n^2 a_0$ for a Rydberg atom of principal quantum number n. An important fact to note about Eq. (15.30) is that $\sigma_m(E_{\mathrm{mw}})$ is proportional to $J_m^2(KE_{\mathrm{mw}}/\omega)$. Using the fact that for small arguments $J_m(x) = (x/2)^m$,[12] at low microwave fields

$$\sigma_m \propto E_{\mathrm{mw}}^{2m}$$
$$\propto I_{\mathrm{mw}}^m, \tag{15.31}$$

i.e. the cross section for an m-photon assisted collision is proportional to the microwave intensity to the mth power, as expected on the basis of time dependent perturbation theory.

The low frequency regime

In the previous discussion of radiatively assisted collisions we have assumed that there are many cycles of the radiation field during the time of one collision, i.e. $\omega/2\pi \gg 1/\tau$. It is interesting to consider the opposite extreme, $\omega/2\pi \ll 1/\tau$. We consider collisions in the total field

$$E = E_{\mathrm{S}} + E_{\mathrm{rf}} \cos(\omega t + \phi), \tag{15.32}$$

and we assume that the collisional resonances are observed by scanning the static field. If, in the absence of the rf field, the resonance is found at $E_S = E_R$, it is found at

$$E_S = E_R - E_{rf}\cos(\omega t + \phi) \tag{15.33}$$

with the added rf field of Eq. (15.32). If we can control the phase of the rf field at which the collision occurs the resonance is simply shifted as shown by Eq. (15.33). If we cannot control the phase, either because the exciting laser is not synchronized with the rf field, or because collisions can occur randomly over several rf cycles, then the probability of observing the resonant collision signal at any given static field is proportional to the likelihood of the total field's being equal to E_R at that static field. If the resonant collision cross sections is given as a function of field by $\sigma_R(E)$, then in the field $E_S + E_{rf}\cos\omega t$ the cross section observed at E_S is given by the convolution of $\sigma_R(E)$ with the function

$$f(E) = \begin{cases} \dfrac{1}{\pi E_{rf}[1-(E-E_S/E_{rf})^2]^{1/2}} & |E-E_S| \leqslant E_{rf} \\ 0 & |E-E_S| > E_{rf} \end{cases} \tag{15.34}$$

In other words $\sigma(E_s)$ is given by the time weighted average of the cross section in the static and oscillating fields. Since the oscillating field spends most of its time at its turning points, the observed cross section can be expected to exhibit the behavior shown in Fig. 15.7.[3]

The intermediate frequency regime

We have considered the high frequency regime, which is independent of the rf phase, and the low frequency regime which is critically dependent on the rf phase. We now consider the intermediate frequency regime, $\omega/2\pi \sim 1/\tau$, approaching it from the low frequency side. To date the only way of calculating the lineshape of the collisional resonances has been explicit numerical integration. As we shall see, when comparisons with experimental results are made, the calculations predict multiple peaks in the cross section. The separation between peaks does not, however, correspond to the rf frequency. We can use a simple picture to understand the origin of the peaks and their locations, although the picture does not give lineshapes.[14]

For concreteness we assume that the collision occurs over the time interval $-T/2 < t < T/2$ in the presence of a rf field of well defined phase so that the total field is given by Eq. (15.32). We consider the transitions between the two molecular states A and B which are degenerate (at $r = \infty$) at the static field E_R. We may assume with no loss of generality that, in the vicinity of E_R, B has no Stark shift and A has a linear Stark shift. Ignoring the dipole–dipole coupling, the energy,

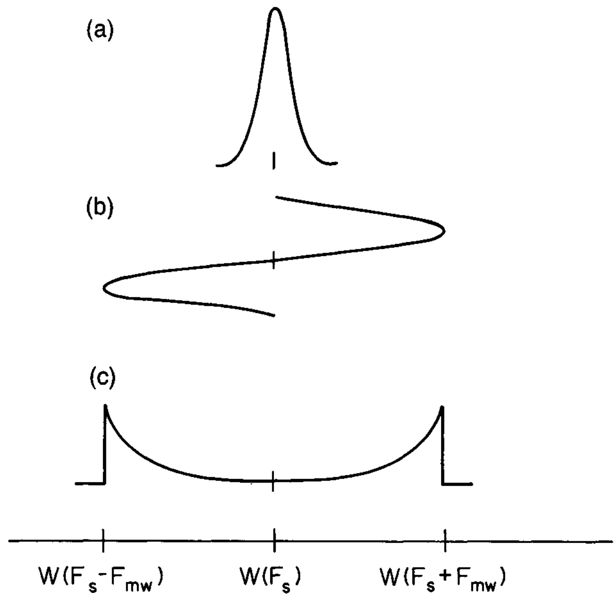

Fig. 15.7 Collisions in the presence of a low frequency microwave field: (a) the resonant collision profile without the microwaves; (b) one period of microwave field showing both the amplitude and the time weighting of the extreme field values; (c) typical time weighted radiative collision lineshape when the microwave period becomes long compared to the duration of one collision (from ref. 3).

W_B, of molecular state B is a constant, and the energy of molecular state A is given by

$$W_A = W_B + k_A(E - E_R), \tag{15.35}$$

where E is given by Eq. (15.32). Inspecting Eq. (15.35) we see that W_A follows the rf field. Imagine that we choose the rf phase $\phi = 0$ so that W_A reaches a maximum in the center of the allowed collision time. In Figs. 15.8(a) and (b), we show the energies W_A and W_B as functions of time for two choices of the phase of a 0.75 MHz rf field, $\phi = 0$ and $\pi/2$, respectively. The allowed collision time $T = 0.7\,\mu$s. The dipole–dipole interaction lifts the degeneracy at resonance, leading to the avoided level crossings shown by the insets of Fig. 15.8. In Fig. 15.8(a) the energies are shown for two values of the static field E_S. For $E_S < E_R - E_{rf}$, the two levels never come into resonance. When $E_S = E_R - E_{rf}$ the two levels are resonant at $t = 0$, as shown by the broken line, and a broad resonance can be expected. For larger E_S the two levels come into resonance twice, as shown by the solid line, and the transition amplitude is the coherent sum of the amplitudes acquired at the two times the levels become resonant, at $t = \pm t'$.

As shown by Fig. 15.8(a), if the system is initially in state A it traverses two avoided level crossings to reach the final state B, and there are two possible paths to go from A to B. Whether the transition amplitudes for these two paths add

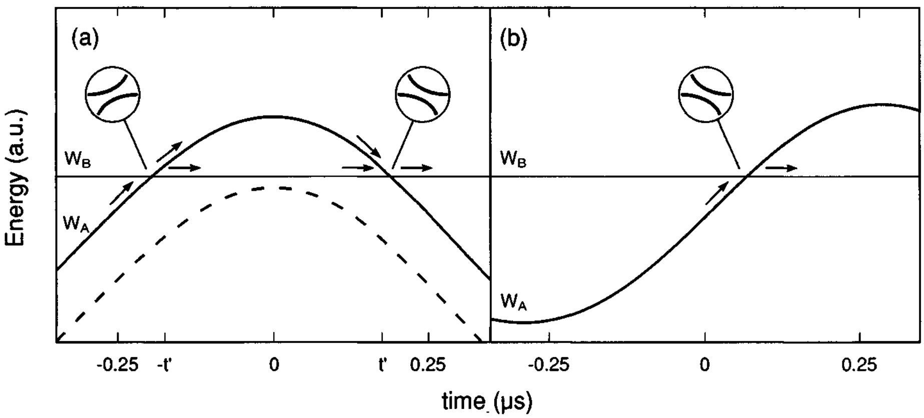

Fig. 15.8 The initial and final state energies W_A and W_B in the total electric field $E = E_S + E_{rf} \cos(\omega t + \phi)$ where the frequency $\omega/2\pi = 0.75$ MHz field and the allowed collision time $T = 0.7\,\mu$s. W_B is field independent and W_A varies linearly with the field. (a) The phase $\phi = 0$. No resonance occurs for $E_S < E_R - E_{rf}$. As the static field is increased, when $E_S = E_R - E_{rf}$ (- - -) the two levels come into resonance at $t = 0$. When $E_S > E_R - E_{rf}$ (—) the two levels come into resonance twice, and there are two resonant interaction periods at $t = \pm t'$. The levels do not actually cross but have avoided crossings as shown by the insets. There are two paths by which atoms initially in state A can reach state B, and they interfere constructively when Φ_B, the phase difference accumulated between $-t'$ and t', is $2\pi N$. In (b) where $\phi = \pi/2$ and $E_S \approx E_R$, only one interaction period occurs for all values of E_S between $E_R - E_{rf}$ and $E_R + E_{rf}$ (after ref. 14).

contructively or destructively depends on the phase difference Φ between the two paths. Φ is simply the area between the two curves in Fig. 15.8(a), i.e.,

$$\Phi = \int_{-t'}^{t'} (W_A - W_B)\,dt. \tag{15.36}$$

If $\Phi = 2\pi N$, where N is an integer the interference is constructive, whereas if $\Phi = 2\pi(N + 1/2)$ it is destructive. As E_S is raised Φ passes through even and odd multiples of π leading to constructive and destructive interference in the cross section. Inspecting Fig. 15.8(a) we can see that the fringes should become closer together as E_S is increased. This description of the origin of the oscillations in the cross section makes it apparent that they are a form of Stückelberg oscillations observed in differential scattering.[15]

The two resonant interactions at $t = \pm t'$ can also be thought of as being analogous to the two separated oscillating field regions in a Ramsey magnetic resonance experiement, in which case the oscillations in the cross section can be thought of as Ramsey fringes.[16]

For $\phi = 0$, as shown in Fig. 15.8(a) the oscillations in the cross sections should extend up to the static field

$$E_S = E_R - E_{rf} \cos(\omega T/2). \tag{15.37}$$

With the rf phase $\phi = \pi$, the collisional resonance is reversed with respect to E_R. With $\phi = \pi/2$, as shown in Fig. 15.8(b) for $E_S \approx E_R$, the result is qualitatively different. The two levels come into resonance at most once for any value of E_S, and as a result there is one broad collisional resonance extending from $E_S = E_R - E_{rf}\sin(\omega T/2)$ to $E_S = E_R + E_{rf}\sin(\omega T/2)$.

The above description is obviously an extension of the low frequency description, but it can also be connected to the high frequency limit. Consider the case in which $\omega \gg 1/T$, i.e., there are many cycles of the field, and $|E_S - E_R| \ll E_{rf}$, the static tuning field is near resonance compared to the rf field amplitude. For a large transition probability the transition amplitudes of successive rf cycles must add constructively. This only happens when the phase difference Φ over an rf cycle satisfies $\Phi = 2\pi N$ with N an integer. Using the energies of Eq. (15.35),

$$\Phi = \int_{-\pi/\omega}^{\pi/\omega} k_A(E_S + E_{rf}\cos\omega t - E_R)\,\mathrm{d}t = \frac{2\pi k_A(E_S - E_R)}{\omega}. \tag{15.38}$$

Equivalently,

$$k_A(E_S - E_R) = N\omega. \tag{15.39}$$

In other words the resonances are detuned from E_R by the static field shift equivalent to N photons, the same result as obtained in the dressed state picture.

Eq. (15.39) tells us where the collisional resonances occur, but it does not tell us how strong they are. The strength is determined by the phase contributions of the two parts of the rf cycle, when $W_A < W_B$ and $W_A > W_B$. When the phases accumulated in these two half cycles are both integral multiples of 2π the collisional resonances are strong. This requirement leads to an oscillation in the intensity of the N photon assisted collisional resonance proportional to $\cos(k_A E_{rf}/\omega + \gamma)$, where γ is a small constant, which has the same period as the Bessel function expression of Eq. (15.29).

While the experiments with Na atoms have validated many aspects of the theory of radiatively assisted collisions in the strong field, high frequency regime, experiments with velocity selected K atoms have extended the measurements both to higher order processes and through the regime in which the collision time matches the frequency of the oscillating field.[14,17,18] Specifically, studying the process

$$\mathrm{K}(n+2)\mathrm{s} + \mathrm{K}\,n\mathrm{d} \rightarrow \mathrm{K}\,n\mathrm{p} + \mathrm{K}\ (n-1)\mathrm{p} + m\omega \tag{15.40}$$

with velocity selected atoms has several attractive features. First, since the collisional resonances are so narrow, it is straightforward to satisfy the condition $\omega/2\pi \gg 1/\tau$ yet still observe collisions radiatively assisted by many photons without exceeding the available tuning range. Second, since the collision times are long, it is possible to observe the predicted phase dependence of the collisions in the transition region $\tau \sim 2\pi/\omega$.

The techniques employed are the same as those used to study the K resonant collisions described in the previous chapter. The apparatus shown in Fig. 14.13 is used, the only difference from the previous experiment being the application of a rf voltage to the top plate of the interaction region. The rf field, at frequencies from 0.5 MHz to 4.0 MHz is either left free running or phase locked to the laser excitation.

Most of the data are acquired in the same way as described above. K atoms are excited to the 29s and 27d states by the laser excitation, 4s $\rightarrow$ 4p $\rightarrow$ 29s, 27d. The atoms are allowed to collide for 1 μs, after which a rapidly rising detuning pulse is applied, followed by the more slowly rising field ionization pulse. Atoms which have made the transition to the 29p state are selectively ionized by the field ionization pulse and detected. This signal is monitored as the small static tuning field is scanned. The amplitude and phase of the rf field are changed as parameters.

Using such low collision frequencies it is possible to study radiative collision processes of high order, a point made explicitly by Fig. 15.9. In Fig. 15.9 we show the development of the radiative collision signal for collisions in the presence of 4 MHz rf fields of amplitudes from 0 to 0.76 V/cm.[17] Unlike the Na case described above, there is only a single collisional resonance when there is no rf field, as shown by Fig. 15.9(a). As the rf field amplitude is raised, progressively higher order resonant collision processes are observed, out to the seven photon assisted collisions. It is interesting to note that both $m > 0$ and $m < 0$ radiatively assisted collisions are observed, approximately symmetrically about the position of the collisional resonance at 6.44 V/cm. A slight asymmetry is expected at the extreme static tuning fields due to the second order Stark shift.

Close inspection of Fig. 15.9 suggests that the strength of a particular m-photon assisted cross section oscillates with the rf field as implied by the Bessel function dependence of Eq. (15.29). To demonstrate this point explicitly, the static field was fixed at the fields corresponding to several m photon assisted resonances and the amplitude of the rf field swept. Following this procedure for $m = 0, +1, -2$, and $+3$ leads to the results shown in Fig. 15.10, which shows quite clearly several zeroes in each of the σ_m cross sections. Furthermore, the experimental results agree almost perfectly with the expression of Eq. (15.28) and the dependence obtained by simply numerically integrating Eq. (14.16) using time dependent energies W_A and W_B. Close examination of Fig. 15.10 reveals that the observed cross sections at the highest rf field amplitudes are smaller than the calculated cross sections. This discrepancy is attributed to the fact that the static fields at which the collisional resonances occur shift slightly due to the second order ac Stark shift as the rf field is increased.

A final aspect of these radiation collisions, shown experimentally by Thomson *et al.*,[17] is that the cross section integrated over all the collisional resonances increases with the rf field. Since the mth collisional resonance has a cross section of $\sigma = \sigma_R|J_m(KE_{mw}/\omega)|$ and $\Sigma_m J_m^2(x) = 1$, in general, $\Sigma\, \sigma_m > \sigma_R$.[17]

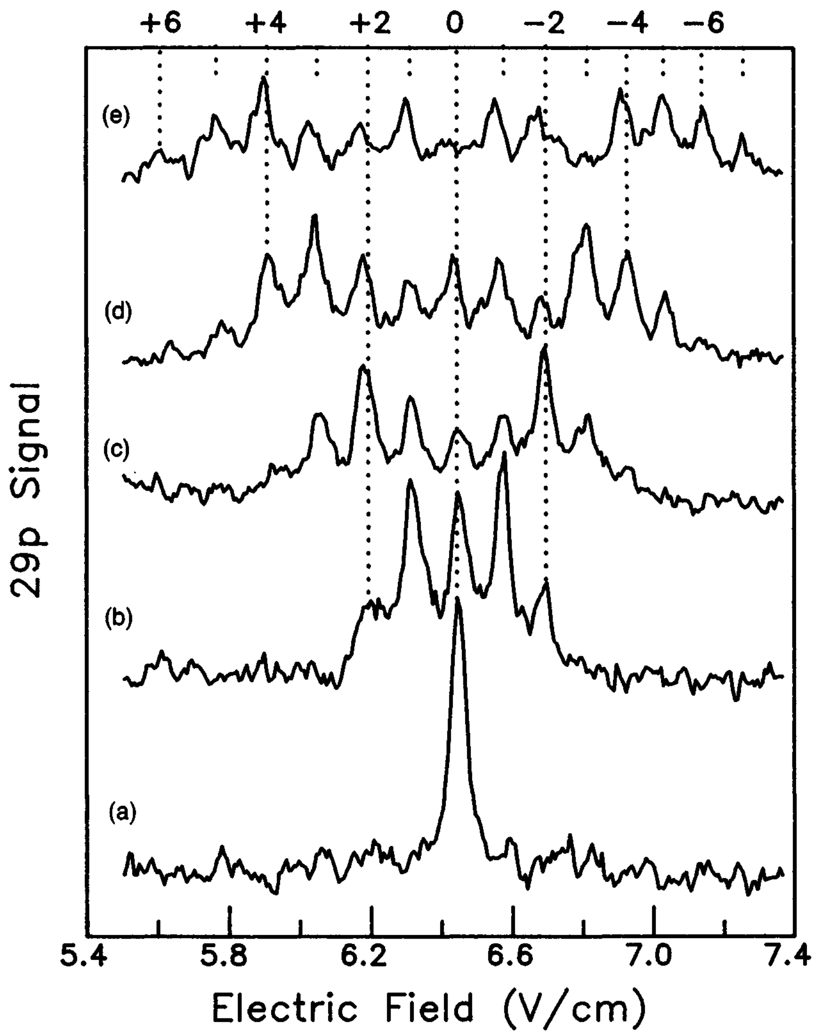

Fig. 15.9 The $n = 29$ resonance of K in the presence of a 4 MHz rf field. The rf field amplitudes for the five sets of data are: (a) 0 V/cm, (b) 0.19 V/cm, (c) 0.38 V/cm, (d) 0.57 V/cm, and (e) 0.76 V/cm. The top axis labels the sideband resonances according to the number of rf photons emitted by the atomic system (from ref. 17).

As we pointed out above, an attractive feature of the velocity selected K collisions is that we can examine the region $\omega/2\pi < 1/\tau$. We consider first the case in which there is no control over the phase of the field at which the collisions occur. This case is exemplified by Figs. 15.11(b)–(d), which show the effect of adding progressively larger amplitude 1 MHz rf fields of uncontrolled phase. Since the laser fires at an uncontrolled phase of the rf field, the observed cross section can be calculated using Eq. (15.34). As shown by Fig. 15.11 the cross section is broadened in approximately the manner shown in Fig. 15.7. As shown by a

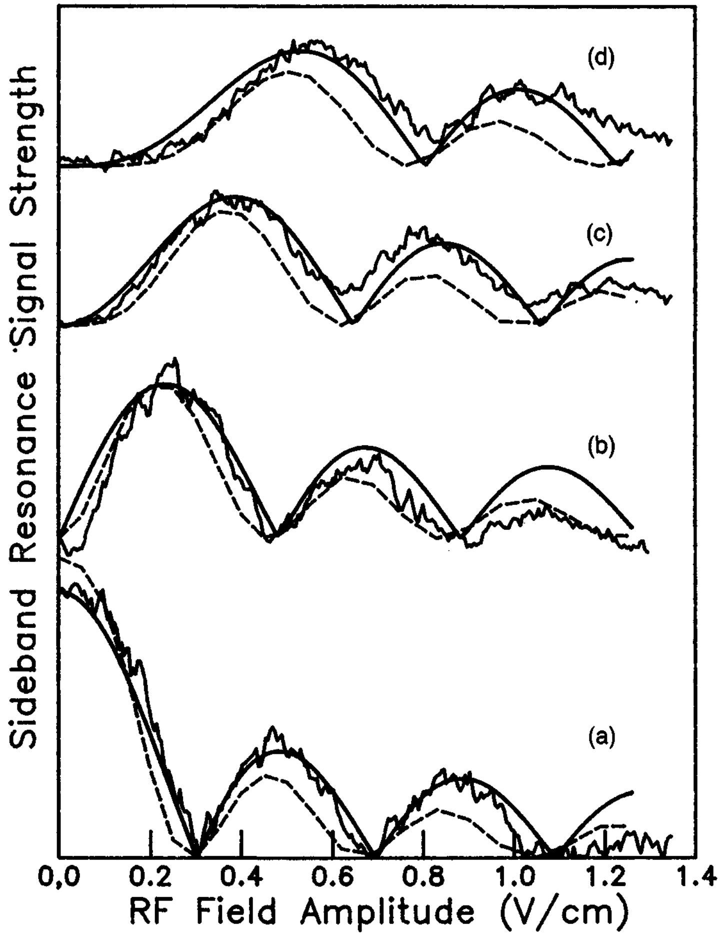

Fig. 15.10 Cross sections as a function of rf field strength for the first four orders of sideband resonances of the K 29s + K 27d radiative collisions in a 4 MHz rf field. (a) The zero-photon resonant collision cross section, (b) the +1 sideband resonance, (c) the −2 sideband resonance, and (d) the +3 sideband resonance. The solid line shows the experimental data, the bold line indicates the prediction the Floquet theory, and the dashed line is the result of numerical integration of the transition probability (from ref. 17).

comparison of Figs. 15.11(d) and 15.11(e), obtained with 1 and 0.5 MHz 0.2 V/cm rf fields, respectively, the broadening of the collisional resonance is independent of the frequency of the applied field, depending only on its amplitude, as predicted by Eq. (15.34).[18]

When the phase of a low frequency rf field is controlled the observed collisional resonances change dramatically. With the field of Eq. (15.32), when the atoms are allowed to collide during the interval $-0.5\ \mu s < t < 0.5\ \mu s$ with $\phi = 0$ or π, we observe the collisional resonances shown in Figs. 15.12(b) and (c) respectively, in agreement with Eq. (15.33). With no rf field the resonance occurs at $E_S = 6.44$

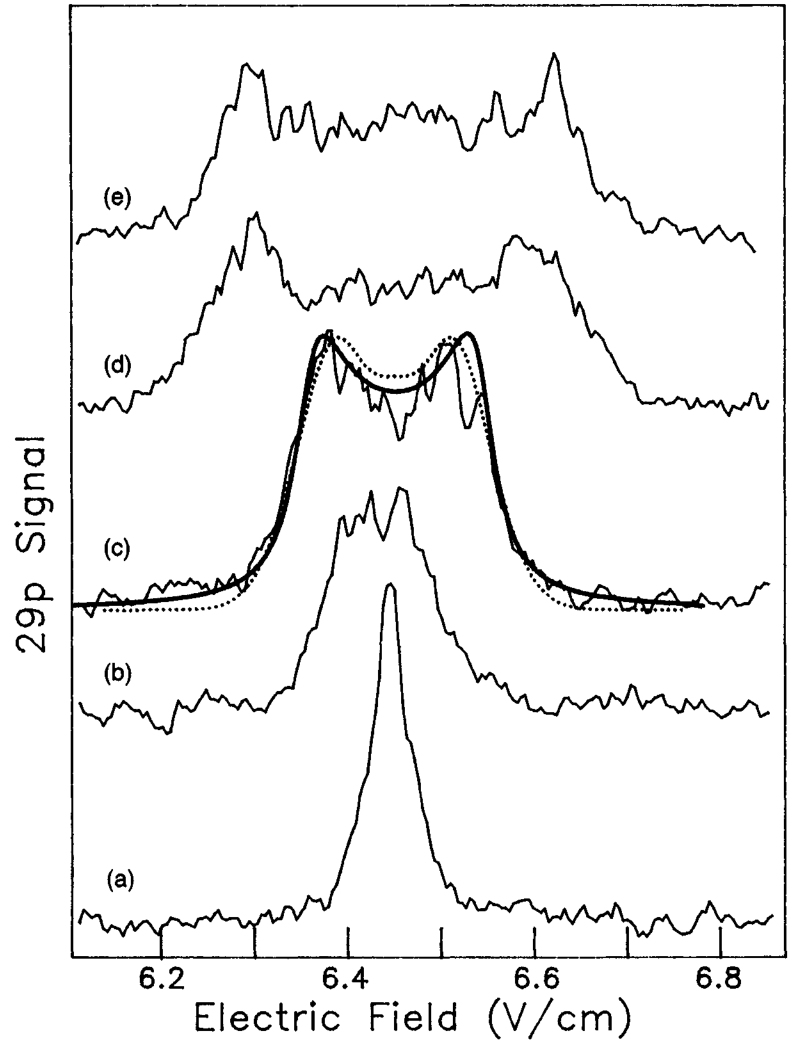

Fig. 15.11 The K 29s + 27d resonance in the presence of a low frequency rf field. In zero rf field (a), the FWHM is 1.6 MHz. In (b)–(d), a 1.0 MHz field of strength 0.05 V/cm, 0.1 V/cm, and 0.2 V/cm respectively is present. The solid line in (b) is a numerical integration of the transition probability, and the bold line is the convolution of a Lorentzian lineshape with a sinusoidal shift from resonance. In (e), the rf frequency is 0.5 MHz and its strength is 0.2 V/cm. For these low frequencies, the features are no long frequency dependent but rather are field strength dependent (from ref. 18).

V/cm, as shown by Fig. 15.12(a). When $\phi=0$ the collision occurs at the maximum of the rf field, and the resonance occurs at a lower static field, as shown in Fig. 15.12(b). Similarly when the collision occurs at the minimum of the rf field the resonance occurs at a higher static field, as shown in Fig. 15.12(c).

Finally we consider the case in which $\omega/2\pi \approx 1/\tau$. In Fig. 15.13 we show the resonances obtained with an allowed collision time $T = 0.8\,\mu$s (the collisions are

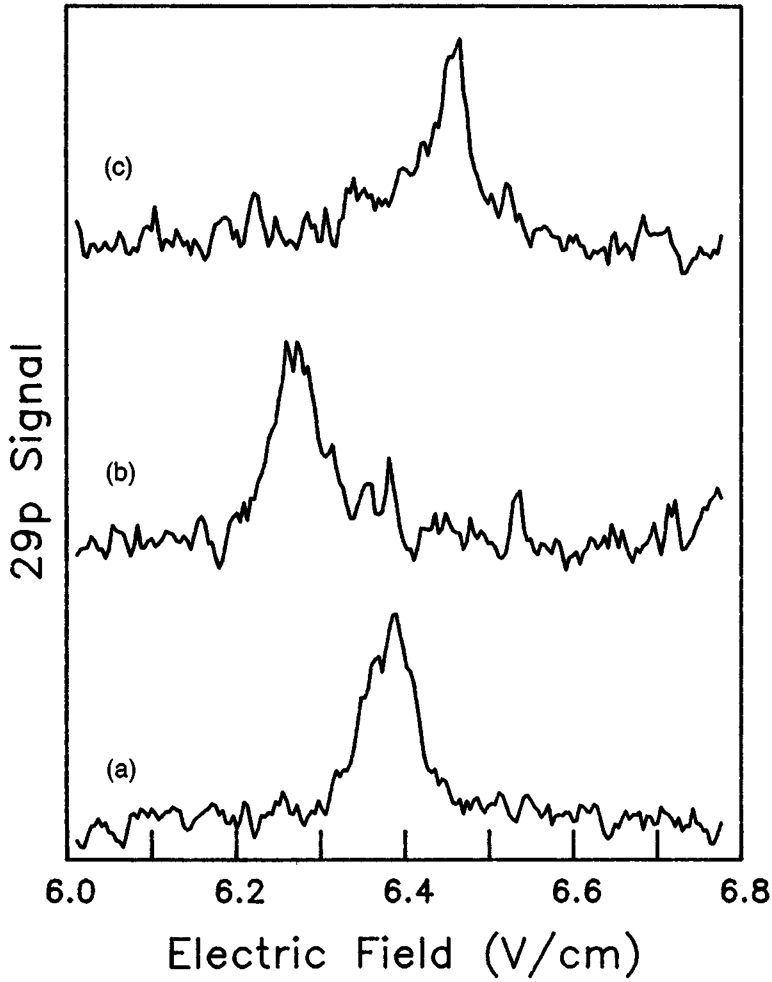

Fig. 15.12 (a) The K 29s + K 27d resonance with a FWHM of 2.0 MHz. In (b) and (c), a 0.5 MHz, 0.1 V/cm rf field is present with a phase of 0 and π respectively. A phase of 0 means the cosine wave is at maximum during the temporal center of the collision (from ref. 18).

allowed to occur for $-0.4\,\mu s < t < 0.4\,\mu s$ and two choices of the phase of an rf field, $E_{rf}\cos(\omega t + \phi)$, of amplitude $E_{rf} = 0.21$ V/cm and frequency 0.75 MHz.[14] In Fig. 15.13(a) we show the resonance with no rf field at 6.44 V/cm. In Fig. 15.13(b), in which $\phi = 0$, the experimental curve exhibits a broad peak at $E_S = 6.27$ V/cm, a subsidiary peak at 6.35 V/cm, and a possible third subsidiary peak at 6.40 V/cm. The dotted lines are numerical calculcations and the locations of the peaks for $N =$ 0–2 from Eq. (15.36) are shown as well. As can be seen, the locations of the experimental peaks agree well with both the numerical calculation and the interference picture of Fig. 15.8(a) which leads to Eq. (15.36). When $\phi = \pi/2$, as shown in Fig. 15.13(c), a single broad resonance is observed, with no apparent

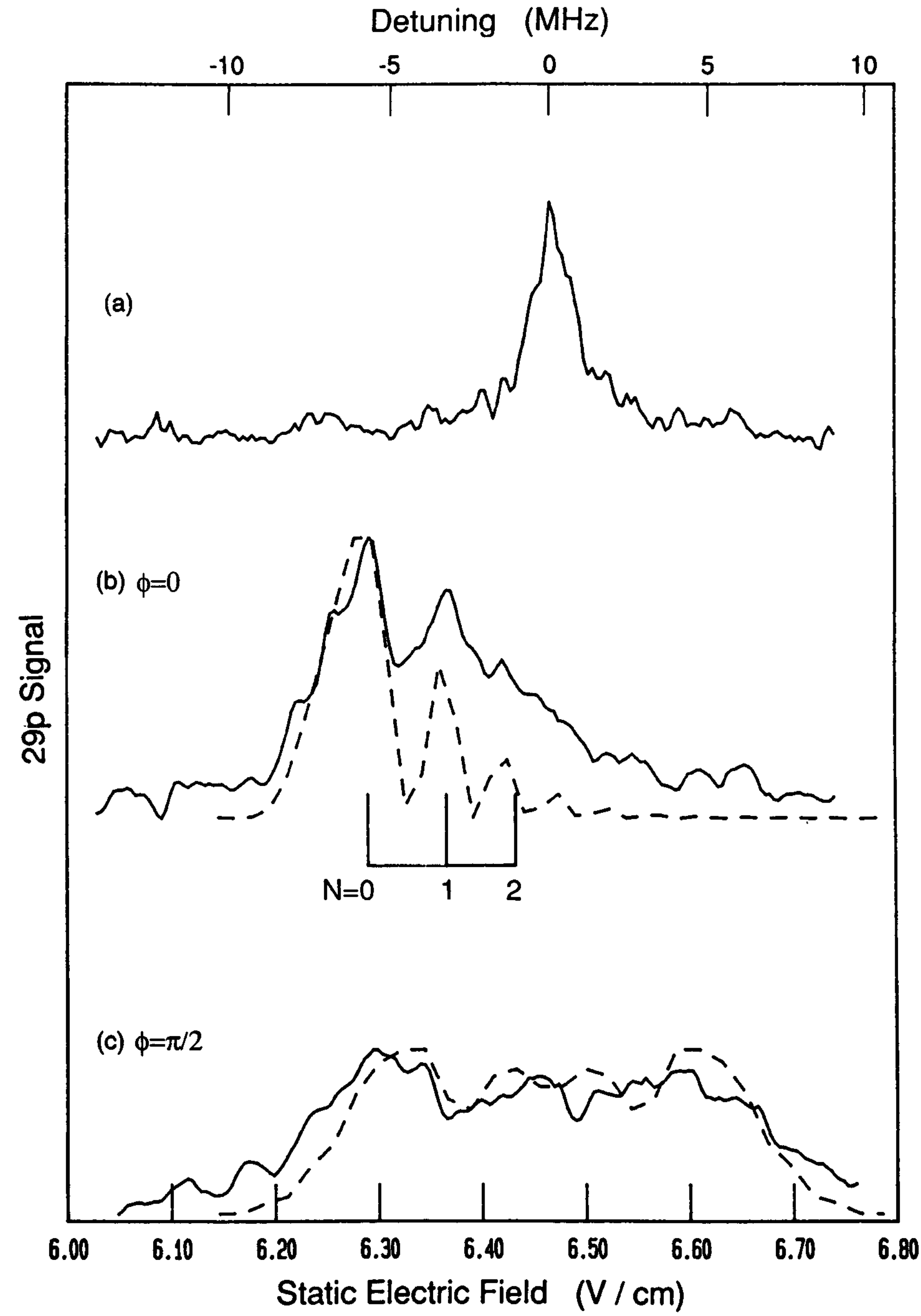

Fig. 15.13 The experimental (—) and calculated (– – –) lineshapes of the n=29 radiatively assisted resonance of K as functions of the static field E_S. (a) The unperturbed resonance with no rf field present, FWHM = 1.4 MHz. (b), (c) The resonance in the presence of a 0.75 MHz, 0.21 V/cm rf electric field and a phase of (b) $\phi = 0$ and (c) $\phi = \pi/2$. Indicated below the spectra are the calculated positions of the peaks using the $\Phi = 2\pi N$ condition (from ref. 14).

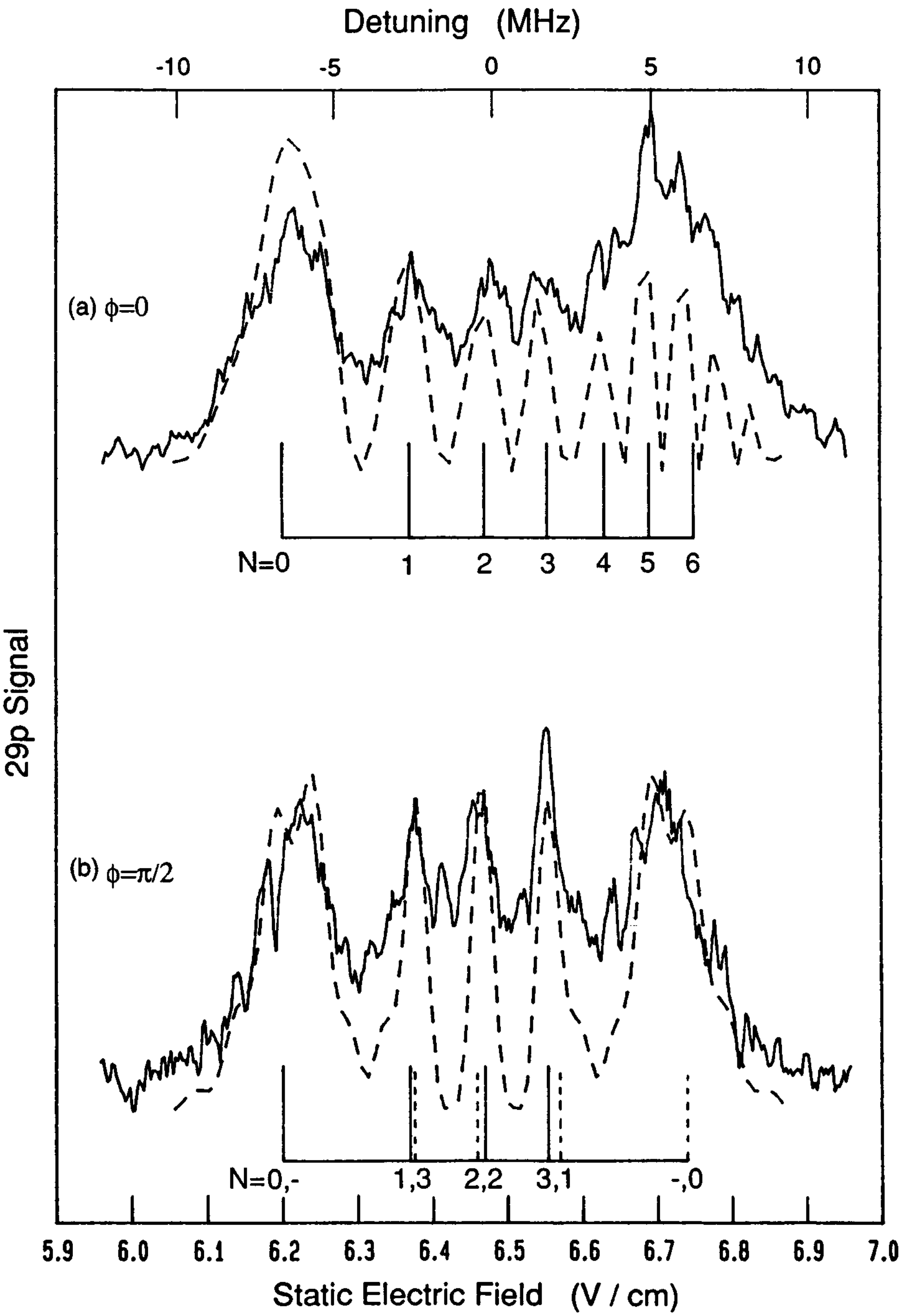

Fig. 15.14 The experimental (—) and calculated (- - -) lineshapes as functions of the static field E_S in a 0.31 V/cm, 1.48 MHz, rf field of phases (a) $\phi = 0$ and (b) $\phi = \pi/2$. The allowed collision time is 0.8 μs. The calculated fringe positions using $\Phi = 2\pi N$ are indicated below the spectra. In (b) the fringe positions due to interference between the first and second (solid vertical marker) and the second and third (dashed marker) interactions are shown (from ref. 14).

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subsidiary peaks. As shown, the numerical calculations agree with the experimental result. Since the levels are only degenerate once, there is no possibility of interference, as shown by Fig. 15.8(b).

As the rf frequency is raised, interference becomes possible for all phases, a point made by Figs. 15.14(a) and (b), showing the results for a 1.48 MHz field of amplitude 0.31 V/cm and phases $\phi = 0$ and $\pi/2$, respectively. The result for $\phi = \pi$ is the mirror image of the $\phi = 0$ result. As shown, the experimental data match the numerical calculation and the prediction of Eq. (15.36).

In addition to the fact that the $\phi = \pi/2$ trace shows interference peaks, we note that they occur at approximately the same static field as the $\phi = 0$ peaks, although they are narrower. As the rf frequency is raised, the phase dependence slowly disappears, until at 4 MHz it is undetectable, and we are in the high frequency regime described by the dressed state picture.

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